

Henry Zhang

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EXPERIENCE

U.S. Department of the Treasury

June 2026 – Present

Software Engineer

Washington, DC

- Architected an event-driven mailbox triage system on AWS with webhook ingestion, failure reconciliation, AI classification, and information extraction, saving ~10 staff-hours per week while automating leadership reporting.
- Unified ~20K financial records weekly across disparate sources using entity deduplication and a shared ontology, and built graph-based relationship models to surface suspicious transactions and hidden entity connections.
- Built an AI security evaluation platform that ran reproducible model evaluations against standardized benchmarks in isolated Docker sandboxes and parallel GCP VMs to assess model exploitability.
- Built and productionized a Microsoft Graph-backed MCP integration for Treasury's internal ChatGPT ecosystem, implementing per-user authentication, data isolation, and request throttling under a 4 req/s service constraint for 4,000+ staff.

Trianz Inc.

September 2025 – June 2026

Associate Project Manager

Herndon, VA

- Managed migration readiness for 60+ applications moving to AWS, translating technical dependencies and deployment blockers into migration plans across Engineering, DevOps, security, and client stakeholders.
- Identified manual reporting bottlenecks and built a full-stack AI application that automated metrics from Excel project trackers and used LLMs to extract risks, blockers, and status updates from unstructured meeting notes, enabling on-demand reporting.

Medidata Solutions Inc.

June 2024 – August 2024

AI Software Engineering Intern

New York, NY

- Built a retrieval-augmented data pipeline combining embedding-based similarity search with keyword filtering to enable high-recall retrieval across 40,000+ medical records.
- Established an evaluation framework for the RAG system using precision, recall, and AI-assisted relevance scoring to benchmark retrieval accuracy, coverage, and trustworthiness
- Optimized PostgreSQL hybrid search by filtering candidates with full-text search before PGvector similarity ranking and reducing unnecessary joins and intermediate data transfer.

Microsoft AI Co-Innovation Labs

June 2023 – August 2023

AI Software Engineering Intern

Shanghai, China

- Developed an enterprise LLM chatbot using embeddings, prompt engineering, and agent workflows to enable natural-language querying across internal company datasets, cutting PM reporting cycles from weeks to days.
- Accelerated go/no-go decisions for client AI and Azure initiatives by translating business needs into scoped Azure and machine-learning pilots with defined data requirements, feasibility criteria, and implementation tradeoffs.

PROJECTS

Inyo | Founding Engineer & Technical Lead, Part-Time

April 2026 – June 2026

- Architected and launched an AI-powered SMS matchmaking platform supporting thousands of users, building the event-driven backend, AI conversation and profile-extraction pipelines, admin tooling, and production infrastructure.
- Led three engineers and partnered with the founder to scope MVP priorities, translate product requirements into technical workstreams, and rapidly deliver features while balancing scalability, reliability, and execution speed.

Fine-tuning and Supervised Learning for Stock Prediction | Machine learning, Data

2024

- Increased labeling accuracy on financial news by 9% by fine-tuning LLMs with qLoRA, improving accuracy from 0.75 to 0.84 to provide more accurate insights in market trend prediction

EDUCATION

New York University

New York, NY

BS in Computer Science, Minor in Business

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML/CSS, C/C++

Tools/Libraries/Frameworks: Git, Docker, React, AWS/Azure/GCP, Excel, Postgres, MongoDB, Redis, BI Tools, Pandas/NumPy, Langchain, pyTorch, MERN Stack, Node.js, Flask, Django, **AWS Cloud Practitioner**.

Skills: Documentation, Project Management, Cross-functional Collaboration, Client Communication, Stakeholder Management